Matthew McCall

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About Me

I am a third-year dual major in computer science and computer systems engineering. I am a dedicated team player who led multiple teams in software development. I have a strong passion for developing robust and performant software on embedded devices.

Skills: C++, React, TypeScript, Git, GitHub, MATLAB, KiCad, SolidWorks, LaTeX, MS Word, Excel, PowerPoint

Education

Rensselaer Polytechnic Institute — **Aug 2022 – (Expected) May 2026**
- Bachelor's in Computer Science and Computer Systems Engineering, Class of 2026

Benjamin N. Cardozo High School — **Sep 2018 – Jun 2022**
- High School Diploma with Regents Advanced Designation with Mastery in Mathematics, Class of 2022
- Member of the National Honors Society (ARISTA)

Work Experience

Engineering and Quality Intern at HarcoSemco — **Sep 2024 – Dec 2024, May 2025 – Aug 2025**
- Led the end-to-end development of an asset tracking system using Bluetooth technology. This included designing the network architecture of gateways and tags, performing hardware selection, developing C++, Java and Kotlin software, and deploying the system to track assets within a 12,000 sq. ft area with an accuracy of 8 feet.
- Wrote C++ software for detecting potential toxic gasses using Bosch sensors and sending data over Bluetooth; and developed the corresponding React Typescript dashboard that uses Web Bluetooth for displaying data in real time.

Undergraduate Research Assistant at SCOREC — **Sep 2022 – Aug 2024, Jan 2025 – Present**
- Reduced runtime by 30% by implementing a GPU memory pool in Omega_h, a C++ library for triangle mesh adaptivity.
- Implemented an algorithm in C++ for finding the nearest simplex to a point in an n-dimensional mesh.
- Authored a scientific paper and poster in LaTeX. I presented my work at two poster sessions at RPI.

Mercer X Lab Assistant — **May 2024 – Aug 2024, Jan 2025 – Present**
- Guided students through safety best practices, circuit design, soldering, equipment usage.

Project Experience

Open Algebra Software for Inferring Solutions (OASIS) — **Sep 2023 – Present**
- Currently leading the team for the OASIS project, an open-source C++ library for computer algebra and symbolic manipulation. I used GitHub Issues, Projects, and Pull Requests to manage the project. I also implemented GitHub Actions CI/CD workflows for automated building and unit testing of the project.

Eye Robot – C#, Arduino C++ — **May 2024 – Aug 2024**
- Eye Robot is a robot that assists the visually impaired in navigation. I worked in a team to write C# and C++ software to construct a 3D model of the room using LiDAR and send commands to an onboard ESP32.

Volunteer Experience

Rensselaer Center for Open Source (RCOS) Mentor — **May 2024 – Aug 2024, Jan 2025 – Present**
- Served as an RCOS mentor, providing guidance and technical support for multiple projects.

Volunteer at The Forge, An RPI Student Makerspace — **May 2024 – Aug 2024, Jan 2025 – Present**
- Assisted students with computer aided design and 3D printing

Live Stream Assistant at St Mary Church, Branford CT — **Jun 2022 – Present**
- Routinely aids with executing livestreams of Sunday Mass.